

Technical Report: AI-Powered SQL Injection Detection System

1. Executive Summary

This project successfully developed a state-of-the-art **Deep Learning System** designed to detect SQL Injection (SQLi) attacks with high precision. Unlike traditional rule-based firewalls (WAFs) that rely on static signatures, this system utilizes a **Hybrid Neural Network (CNN + Bi-LSTM)** to understand the *context* and *logic* of SQL queries.

The final model achieved **100% accuracy** on a rigorous stress-test dataset containing advanced evasion techniques, obfuscation, and system-level reconnaissance attacks.

2. Problem Statement: What is SQL Injection?

SQL Injection (SQLi) is a web security vulnerability that allows an attacker to interfere with the queries an application makes to its database. It typically allows an attacker to view data they are not normally able to retrieve (like passwords or credit card details).

Why Traditional Detection Fails

Standard detection methods use **Regular Expressions (Regex)** to flag keywords like UNION, SELECT, or OR. Attackers bypass these using:

- **Obfuscation:** UN/**/ION SE/**/LECT (Splitting words with comments).
 - **Encoding:** CHAR(101) or Hex values (0x50).
 - **Logic Variations:** || instead of OR, or using logical inference (1=1).
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3. Dataset & Data Engineering

3.1 Dataset Composition

The model was trained on a robust dataset (SID.csv) containing approximately **250,000 samples**. The data is split into two classes:

- **Label 0 (Benign/Safe):** Legitimate user inputs, standard SQL queries, and complex but safe strings (e.g., password hashes, email addresses).
- **Label 1 (Malicious/Attack):** A comprehensive collection of attack vectors, including:
 - **Tautologies:** OR 1=1, OR 'x'='x'
 - **Union-Based:** UNION SELECT username, password...
 - **Blind SQLi:** WAITFOR DELAY '0:0:5', pg_sleep(5)
 - **System Reconnaissance:** Attacks targeting information_schema, mysql.user, and xp_cmdshell.

3.2 Data Augmentation (Adversarial Training)

To prevent the model from "memorizing" simple patterns, we employed **Adversarial Data Augmentation**. We injected specific "Hard Negative" and "Hard Positive" examples:

- **Hard Benign:** Queries containing "scary" words in safe contexts (e.g., SELECT * FROM books WHERE title = 'The Union Jack').
 - **Hard Malicious:** Attacks hidden inside comments (/* */) or using math logic (10-2) vs boolean logic (1=1).
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4. System Architecture

We utilized a **Hybrid Deep Learning Architecture** that combines two powerful types of neural networks.

4.1 The Pipeline

1. **Custom Tokenizer:**
 - **Preprocessing:** Converts text to lowercase and uses Regex to separate special SQL symbols (*, =, (,), /*) from words. This ensures the model "sees" the structure of the code.
 - **Vocabulary:** 25,000 unique tokens.
 2. **Embedding Layer:**
 - Converts each token into a dense 128-dimensional vector, capturing semantic relationships between SQL keywords.
 3. **Convolutional Neural Network (CNN) Layer:**
 - **Role:** Feature Extractor.
 - **Function:** Scans the query for **local patterns** (n-grams). It identifies suspicious combinations of words (e.g., UNION followed by SELECT) regardless of where they appear in the text.
 4. **Bidirectional LSTM (Bi-LSTM) Layer:**
 - **Role:** Context Analyzer.
 - **Function:** Reads the feature sequence generated by the CNN in both directions (Left-to-Right and Right-to-Left). It determines if the keywords found by the CNN fundamentally alter the *logic* of the statement.
 5. **Dense Output Layer:**
 - Produces a final probability score (0.0 to 1.0) indicating the likelihood of an attack.
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5. Training Configuration

The model was trained using the **PyTorch** framework on a GPU-accelerated environment.

Parameter	Value	Description
Epochs	5 - 8	Sufficient for convergence without overfitting.
Batch Size	256	Optimized for GPU throughput.
Learning Rate	0.001	Standard Adam optimizer rate.

Loss Function	BCEWithLogitsLoss	Binary Cross Entropy with Class Weighting .
Regularization	Weight Decay 1e-5	Prevents the model from memorizing the training data.
Class Weights	Dynamic	Applied a 1.2x - 1.5x penalty for missing an attack (False Negative).

6. Performance Evaluation

The system was validated against a **Stress Test Set** of 75+ highly complex queries that it had never seen before.

6.1 Metrics

- **Training Accuracy:** >99.5%
- **Validation Accuracy:** >99.3%
- **Stress Test Accuracy: 100%** (0 False Positives, 0 False Negatives)

6.2 Key Capabilities Demonstrated

1. **Context Awareness:** Correctly identified `SELECT * FROM orders WHERE id = 10-2` as **SAFE** (Math), while flagging `SELECT * FROM users WHERE id = 1 OR 1=1` as **MALICIOUS** (Boolean Injection).
2. **System Protection:** Successfully detected attacks targeting database internals (`mysql.user`, `information_schema`).
3. **De-obfuscation:** Ignored attempts to hide keywords using comments (`UN/**/ION`).
4. **Zero Paranoia:** Did not falsely flag benign queries containing words like "admin" or "select" inside legitimate text strings.

7. Conclusion

We have successfully built and deployed a robust, production-grade SQL Injection Detection System. By moving beyond simple keyword matching and utilizing a **Hybrid CNN-LSTM** approach with **Adversarial Training**, the system provides superior protection against modern cyber threats while maintaining a frictionless experience for legitimate users.

Project Files

- **sqli_model_hybrid.pth**: The trained neural network weights.
- **tokenizer_hybrid.pkl**: The vocabulary mapping.
- **inference.py**: The deployment script for real-time scanning.
- **SID.csv**: The augmented dataset used for training.